

UMESH CHANDRA TIRUMANI

+91 6303336036 ✉ thirumaniumeshchandra2006@gmail.com [LinkedIn](#) [Github](#) [Credly](#)

Education

Anurag University, Hyderabad
B.Tech in Computer Science & Engineering

2023 - Present
8.6/10 CGPA

Work Experience

AlgoUniversity (backed by Y-Combinator)

May 2025 – July 2025

SWE Co-op Intern

Remote

- Built and deployed an online judge platform during internship, enabling **50+ students** to practice coding problems daily.
- Developed RESTful APIs in **Node.js/Express.js** to support course management and student submissions, cutting manual operations by 40%.
- Optimized database queries in **MongoDB**, improving data retrieval speed for high-traffic endpoints by 25%.
- Enhanced frontend components with **React.js**, reducing page load times by 20%.
- Deployed services on **AWS, Render, Vercel** and automated builds with **Docker**, ensuring a stable and scalable environment.
- Collaborated with a team of 5 engineers, participated in weekly scrum calls, and managed progress through daily updates in a shared tracking sheet and delivered 3 major features ahead of schedule.

Projects

Code Slayer: Full-Stack Coding Platform | *React.js, Node.js/Express.js, MongoDB* |

[Source Code](#) — [Live Demo](#)

- Developed a full-stack **Online Judge System** leveraging **React.js, Node.js/Express.js**, and **MongoDB**, enabling problem-solving with secure **multi-language code execution** under strict time & memory limits.
- Implemented a real-time **code editor & compiler** (Monaco Editor) with scalable REST APIs for **problem management, user authentication**, and **submission history**.
- Optimized **MongoDB schemas** to handle **10,000+ records** efficiently and integrated **JWT authentication with bcrypt hashing**, reducing login errors by **25%**.
- Containerized the code execution engine with **Docker**, improving evaluation speed by **40%**, and deployed services on **AWS EC2, Render**, and **Vercel** for scalability.
- Integrated an **AI-assisted code review system (Gemini API)** to provide feedback and enhance the learning experience.
- Validated system performance with **500+ test submissions**, achieving average execution time under **2 seconds** for smooth user experience.

Expense Tracker: Personal Finance Manager | *Python, Tkinter, SQLite, Matplotlib* |

[Source Code](#)

- Designed and implemented a **Personal Expense Tracker** as a lightweight Python application with a **Tkinter GUI** for intuitive financial management.
- Enabled users to **add, edit, and categorize** transactions (income and expenses) and set monthly budgets, supporting flexible financial tracking.
- Developed persistent storage and fast retrieval using **SQLite**, with integrated advanced analytics to visualize budget usage and spending breakdowns, utilizing **Matplotlib** for interactive charts and reports.

Technical Skills

Languages: Python, Java, C++, C, JavaScript
Frameworks & Libraries: Node.js, Express.js, React, TailwindCSS
Databases & Cloud: AWS, Render, Vercel, MongoDB, MySQL
DevOps & Tools: Docker, Postman, VS Code, GitHub

Honours and Awards

Accelerator Programming Camp

March 2025 – August 2025

Batch of 2025

AlgoUniversity (backed by Y-Combinator)

- Selected among the top **1.4% out of 40,000** for AlgoUniversity's Accelerator Program, receiving mentorship from senior engineers at Google, Apple, Grab, and Alphagrep.
- Participated in **Graphs & Dynamic Programming Camps**, strengthening algorithmic thinking and competitive programming skills.
- Gained practical experience in building and deploying scalable systems, applying industry-level software development practices.

Achievements/Certifications

- Demonstrated strong algorithmic problem-solving skills on LeetCode (Rating 1497) and earned multiple Badges in Hackerrank.
[Leetcode](#) [Hackerrank](#)
- Participated in 3+ hackathons, building prototypes in teams of 3–5 under 24hr deadlines.
- View verified certifications portfolio: [Google Drive](#)
- View public profile with Google Cloud Boost achievements: [Public Profile](#)